# Ritesh Kumar Chauhan

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## Education

- **B.M.S College of Engineering- Bangalore, India** Expected 2027
  *Bachelor of Engineering - Information Science* *CGPA: 9.6*
- **Patna Doon Public School- Patna, India** May 2023
  *Class 12th, CBSE* *Percentage: 92.4*
- **Don Bosco Academy- Patna,India** July 2021
  *Class 10th, ICSE* *Percentage: 97.4*

## Skills Summary

- **Languages** C++, C, Python, Java, HTML, CSS, JavaScript, SQL, Motoko, JSON, Bash
- **Frameworks and Libraries** FastAPI, Express.js, React, REST APIs, scikit-learn, NumPy, Pandas
- **Developer Tools** Node.js, Git, GitHub, VS Code, DFX SDK, Postman, npm, Docker, AWS
- **Databases** PostgreSQL, MySQL, MongoDB
- **Soft Skills** Teamwork, Communication, Leadership, Event Management

## Projects

- **Autonomous Self-Healing ML System** | *Python, FastAPI, React, Scikit-Learn, River, SciPy, Pytest* Link
  - Built a production-grade self-healing ML pipeline for predictive maintenance using the NASA CMAPSS dataset, continuously monitoring live sensor streams and autonomously retraining degraded models without operator intervention.
  - Implemented multi-layer monitoring with ADWIN concept drift detection, KS-test feature drift detection, Isolation Forest anomaly detection, adaptive rate limiting, and shadow A/B evaluation before promoting improved models.
  - Developed a FastAPI backend with a React dashboard exposing real-time pipeline metrics, audit logs, fault injection scenarios, and model health, validated by a comprehensive 74-test Pytest suite.
- **Decision Autopsy – Multi-Agent Decision Simulation System** | *FastAPI, React, Pydantic, LLM APIs* Link
  - Architected a 5-agent LLM pipeline where each agent processes and forwards structured outputs to the next, enabling sequential multi-stage decision simulation.
  - Built a FastAPI backend with strict Pydantic validation (schema constraints, ordering rules, Literal enforcement) to ensure deterministic, UI-safe agent outputs.
  - Developed an interactive React interface for real-time "what-if" analysis, visualizing 4 simulated future outcomes per decision.
- **Network Intrusion Detection Analysis** | *React, Node.js, Express.js, PostgreSQL, PySpark, Scikit-Learn, Passport.js* Link
  - Processed 2.8M+ CICIDS-2017 records using PySpark, applying feature engineering and scaling to train 7 models achieving 99.86% intrusion-detection accuracy.
  - Built a full-stack IDS dashboard with React 19 + Node.js/Express, featuring PostgreSQL, Passport.js (local + Google OAuth 2.0), and role-based access control for model evaluation.
  - Implemented interactive Recharts visualizations to compare model metrics with dynamic grouped bar charts, enabling real-time analysis of accuracy, precision, recall, and F1-scores.

## Leadership and Extracurricular Activities

- **Junior Coordinator- PhaseShift'24** Led a team of volunteers to organize and execute PhaseShift'24, the college's technical fest. Collaborated with senior coordinators and contributed to overall event planning
- **Volunteer- Utsav'24** Assisted in organizing and executing a college-level esports Valorant tournament, managing team registrations, match scheduling, and real-time coordination during the event. Helped ensure smooth technical setup and maintained fair play and participant engagement throughout the competition.